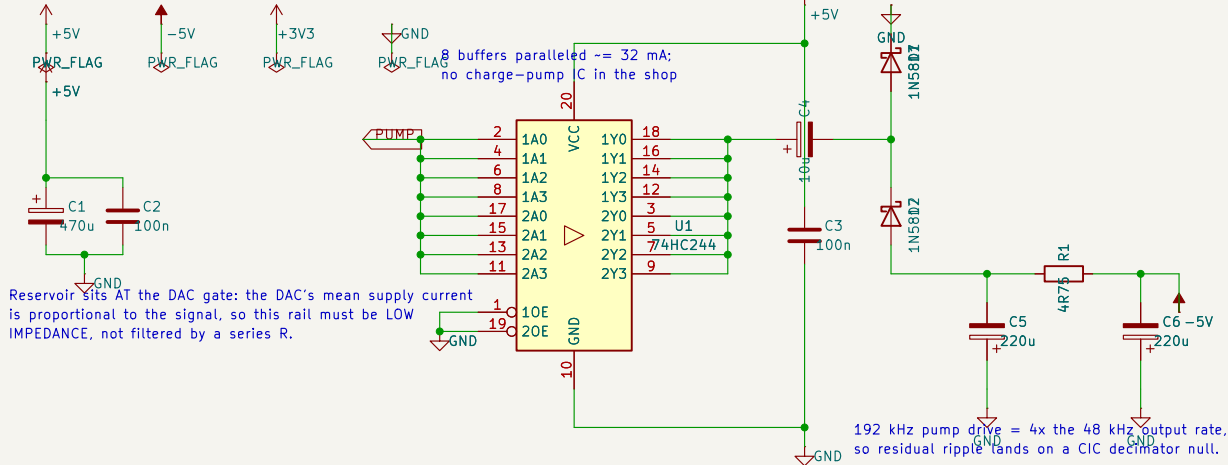


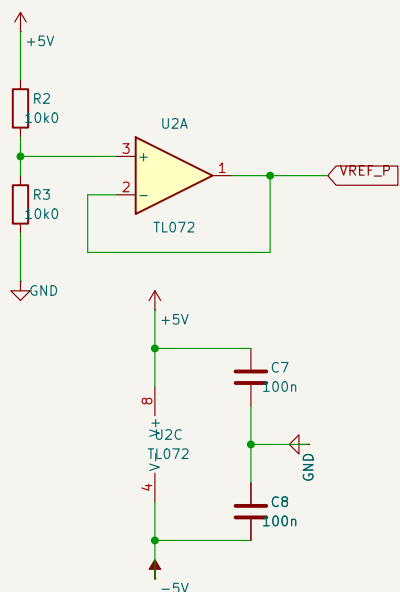
POWER (Pi +5V -> charge-pump -5V -> +/-2.5V DAC reference)



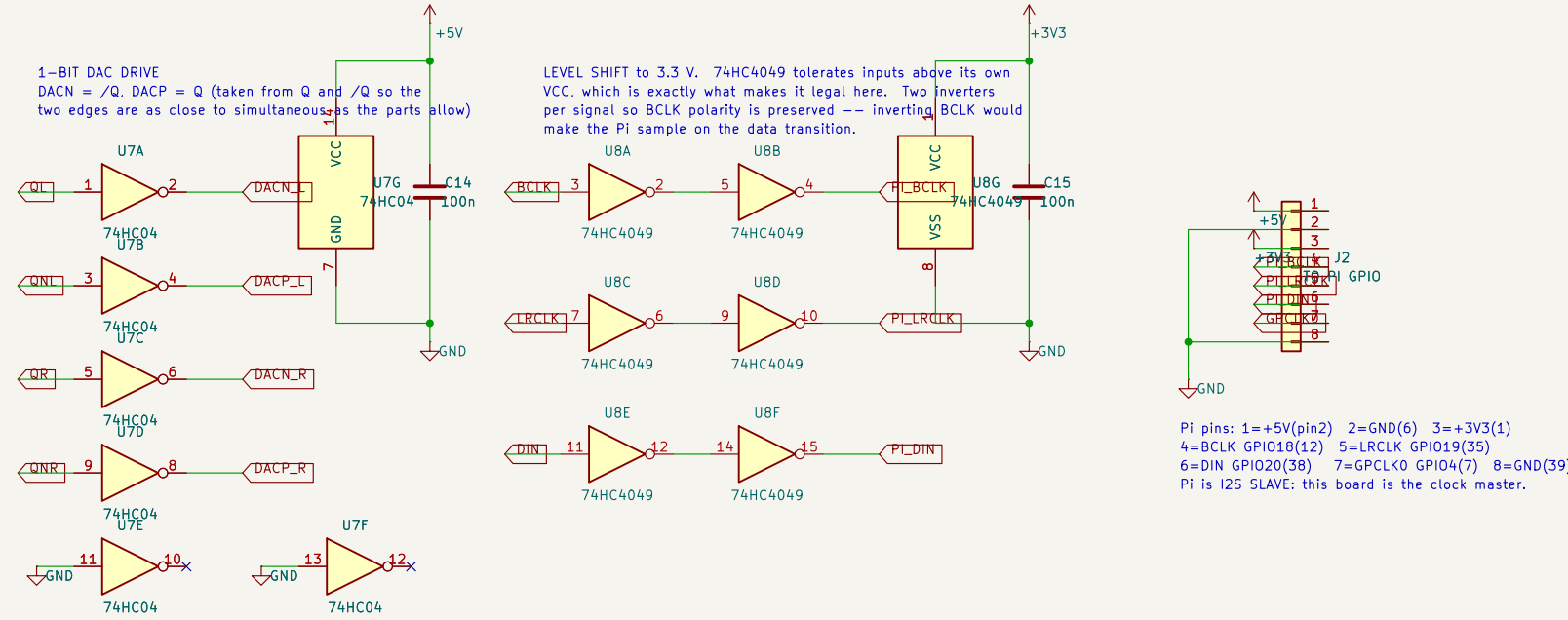
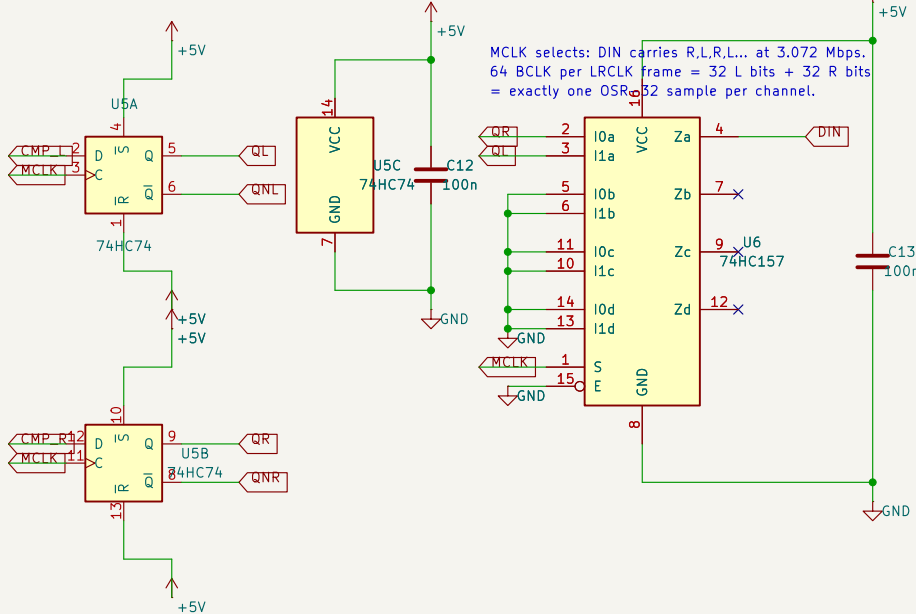
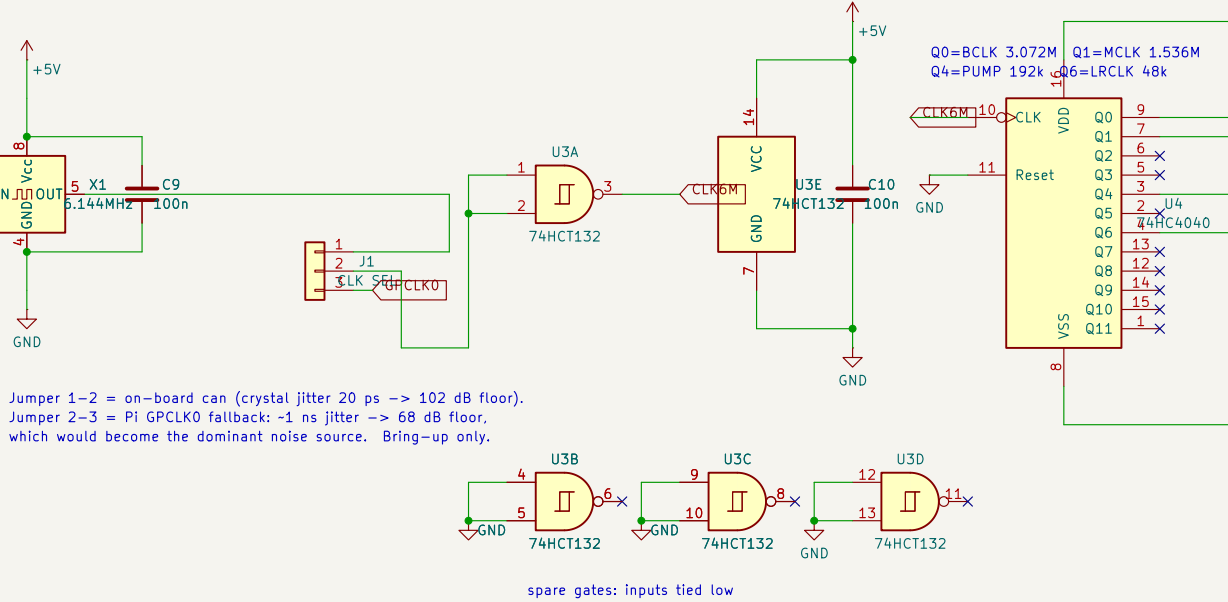
SPICE TESTBENCHES (sim/, one project each -- click to open)

- A rails and the -5 V charge pump
- B the +/-2.5 V ratiometric reference
- C clock buffer and 74HC4040 divider
- D integrator 1 with a real TL074
- E LM311 quantiser on a single supply
- F the 1-bit DAC and its offset leg
- G retiming, interleave and the 3.3 V level shift
- H one complete channel, with the click test

REFERENCE (+/-2.5V, ratiometric -- tracks +5V so rail noise cancels as gain, not additive noise)



CLOCK AND DIGITAL (6.144 MHz -> /2 BCLK 3.072M, /4 MCLK 1.536M, /128 LRCLK 48k; L and R interleaved onto one DIN)



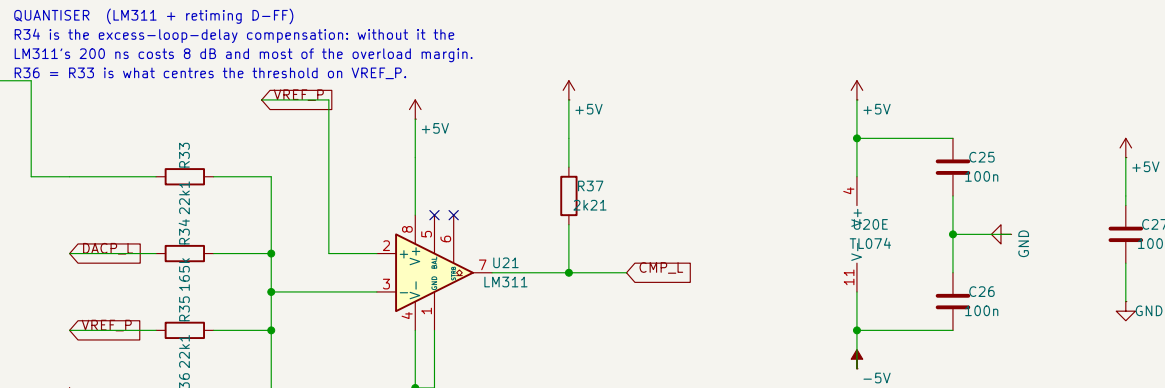
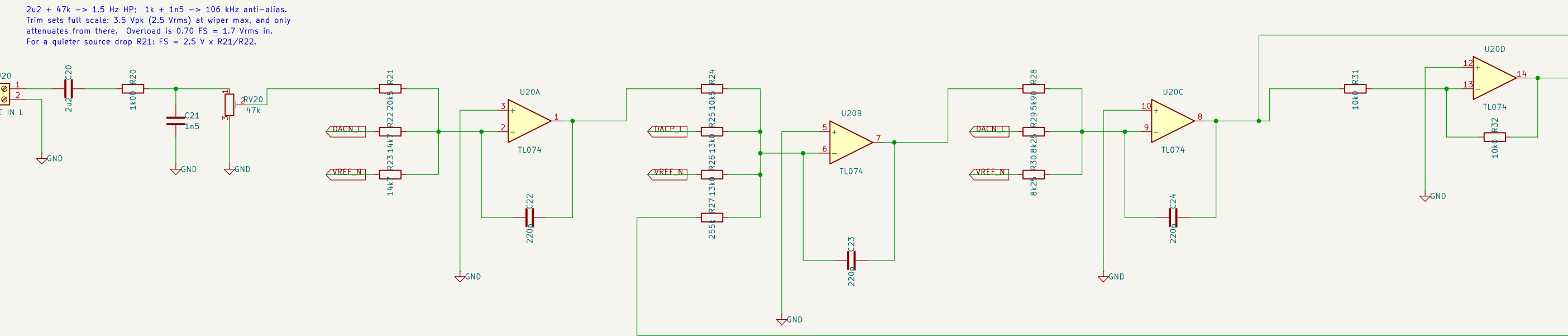
CHANNEL L - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)

INTEGRATOR 2 (a2 = 0.321, + resonator)

INTEGRATOR 3 (a3 = 0.611)

RESONATOR INVERTER (g = 0.0297)



CHANNEL R - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)

INTEGRATOR 2 (a2 = 0.321, + resonator)

INTEGRATOR 3 (a3 = 0.611)

RESONATOR INVERTER (g = 0.0297)

